

Business Report: Scalable E-Commerce Platform Architecture

Prepared for:

Emily Watson, IT Technical Lead
Mphasis (UK)

Prepared by:

Pravin Darade, Associate Architect
Mphasis.

Date:

17 April 2025

Executive Summary

This comprehensive report presents our architectural solution for a scalable e-commerce platform, incorporating UK-specific compliance requirements such as GDPR and Strong Customer Authentication (SCA) to ensure data privacy and secure transactions. designed specifically to handle high-traffic events like Black Friday while addressing UK-specific market considerations. The proposed microservices-based architecture will support 1 million daily active users, 250,000 peak concurrent users, and process up to 10,000 transactions per second across web and mobile platforms.

Based on extensive analysis, we recommend implementing a cloud-native microservices architecture using a UK-based cloud provider such as AWS UK or Azure UK. Leveraging region-specific services—like AWS Lambda, Azure Functions, or regional data centers—ensures adherence to data sovereignty laws and enhances latency performance for UK users. with UK-region deployment, enhanced security measures, and performance optimisations specifically tailored to UK shopping behaviours. This solution ensures compliance with UK regulations while delivering a seamless customer experience during peak traffic periods.

1. Introduction & Business Context

1.1 UK E-Commerce Market Analysis

The UK e-commerce market presents unique characteristics that significantly impact architectural decisions:

- Black Friday in the UK has evolved beyond a single day to become a week-long sales period, creating sustained high traffic rather than a 24-hour spike
- UK consumers show distinct shopping patterns with peak activity between 19:00-22:00 GMT on weekdays and throughout afternoons on weekends
- Mobile commerce represents 73% of UK Black Friday transactions, significantly higher than desktop usage
- UK consumers expect transparent pricing (including VAT) and clear delivery timeframes upfront

- Payment preferences in the UK lean heavily toward debit cards and digital wallets, with growing adoption of Open Banking payments

These patterns necessitate an architecture that can maintain consistent performance over an extended sales period while optimising for UK-specific user behaviours and expectations.

1.2 Technical Requirements

The platform must satisfy both functional and non-functional requirements:

Functional Requirements:

- Secure user authentication with SSO and MFA
- Comprehensive product catalogue and shopping cart functionality
- Checkout process with multiple UK payment gateway integrations
- Order tracking, notifications, and review systems
- Administrative dashboard with analytics and role-based access

Non-Functional Requirements:

- 99.99% uptime with automatic scaling for peak loads
- API response times under 200ms
- Payment success rate exceeding 98%
- Compliance with UK GDPR, PCI DSS, and UK e-commerce regulations
- Real-time monitoring and graceful degradation protocols

2. System Architecture Overview

2.1 Systems and Subsystems

Our proposed architecture comprises several interconnected systems and subsystems, each responsible for specific aspects of the e-commerce platform:

Core Domain Services:

1. Authentication Service

- Handles user registration, login, and account management
- Implements SSO and MFA using UK-trusted identity providers
- Manages role-based access control for system administration
- Stores user credentials with appropriate encryption and hashing

2. Product Catalogue Service

- Manages product information, categories, and search functionality

- Handles pricing logic including VAT calculations for UK customers
- Provides product recommendations based on browsing history
- Maintains inventory status and availability information

3. Shopping Cart Service

- Enables adding/removing products and quantity adjustments
- Calculates totals including accurate UK VAT and shipping costs
- Preserves cart contents across sessions and devices
- Implements cart expiration and recovery mechanisms

4. Checkout Service

- Manages multi-step checkout process
- Validates delivery addresses using UK postcode database
- Integrates with payment service for transaction processing
- Handles order confirmation and receipt generation

5. Payment Service

- Integrates with multiple UK payment gateways (Worldpay, Barclaycard, PayPal)
- Implements payment method selection and validation
- Provides failure handling and retry mechanisms
- Ensures PCI DSS compliance for all payment processing

6. Order Management Service

- Processes orders and maintains order status
- Integrates with UK logistics and shipping providers
- Manages order fulfillment workflows
- Supports order tracking and history

Supporting Services:

7. Notification Service

- Sends transactional emails, SMS, and push notifications
- Manages marketing communications with consent tracking for UK regulations
- Provides notification preferences and opt-out management
- Handles delivery failure and retry logic

8. Review & Rating Service

- Collects and displays customer reviews and ratings
- Implements moderation workflows
- Calculates aggregate ratings
- Prevents review spam and abuse

9. Analytics & Reporting Service

- Gathers performance and business metrics
- Generates real-time dashboards and scheduled reports
- Provides customer behaviour insights
- Supports business intelligence functions

Infrastructure Components:

10. API Gateway

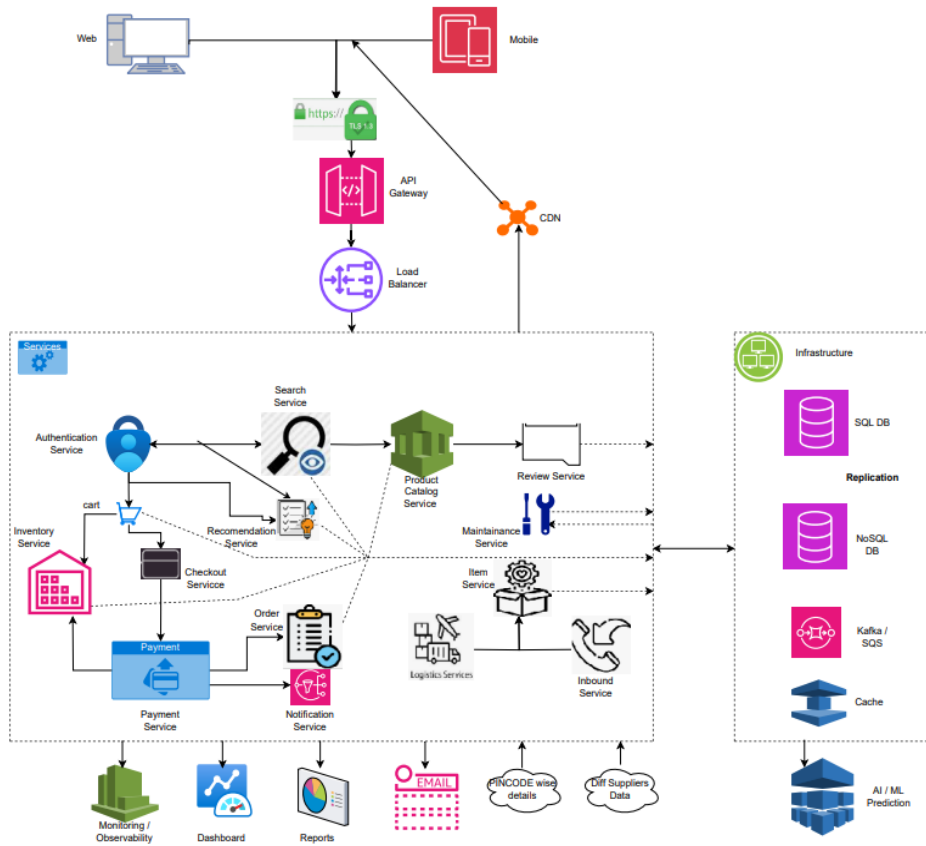
- Routes client requests to appropriate services
- Handles authentication and authorization
- Implements rate limiting and request throttling
- Provides logging and monitoring hooks

11. Content Delivery Network (CDN)

- Caches static content close to UK users
- Reduces load on origin servers
- Provides DDoS protection
- Optimises content delivery based on device and connection

2.2 Architecture Diagram

The following diagram illustrates the high-level architecture of the e-commerce platform:



2.3 Technology Stack

The recommended technology stack consists of:

- **Frontend:** React.js for web, Swift for iOS, Kotlin for Android
- **API Gateway:** Amazon API Gateway with AWS CloudFront CDN
- **Microservices:** Containerized services using Docker and Kubernetes
- **Databases:** Mix of PostgreSQL (transactional) and MongoDB (product catalogue)
- **Caching:** Redis for session data and frequently accessed content
- **Message Broker:** Apache Kafka for event-driven communication
- **Infrastructure:** AWS UK Region (eu-west-2 London) with multi-AZ deployment
- **CI/CD:** Jenkins with GitLab for version control
- **Monitoring:** Prometheus, Grafana, and ELK stack

3. Information Flow and Exchange

To support scalability and resilience, the system will adopt an event-driven messaging architecture leveraging technologies such as Apache Kafka. This approach enables asynchronous communication between services, decoupling producers and consumers to allow for greater flexibility, fault tolerance, and responsiveness under peak loads. Data exchange will rely on lightweight formats such as JSON over REST and gRPC where applicable, with schema validation to ensure consistency. Additionally, critical transactions and state changes will be logged through durable, append-only streams to facilitate replay and recovery mechanisms in case of failures

3.1 Communication Patterns

The platform utilizes two primary communication patterns:

1. Synchronous Communication (REST APIs)
- Used for user-facing operations requiring immediate response
 - Suitable for product browsing, cart updates, and checkout initiation
 - Implemented with RESTful principles and JSON payloads
 - Response time targets of <200ms for critical operations
2. Asynchronous Communication (Event-Driven)
- Used for background operations and inter-service communication
 - Appropriate for order processing, inventory updates, and notifications
 - Implemented using Kafka message streams
 - Ensures system resilience during peak load periods

3.2 Data Exchange Characteristics

The data exchange within the system varies by function:

FUNCTION	FREQUENCY	DATA VOLUME	SPEED REQUIREMENT	UK-SPECIFIC CONSIDERATIONS
PRODUCT BROWSING	Very High	Medium	<200ms	UK pricing format with VAT included
CART UPDATES	High	Low	<100ms	UK shipping options and tax calculations
CHECKOUT	Medium	Medium	<200ms	UK address validation, payment methods
PAYMENT PROCESSING	Medium	Low	<500ms	UK banking integration response times

ORDER MANAGEMENT	Low	High	<1s	Integration with UK shipping providers
ANALYTICS	Low	Very High	Minutes	UK business reporting requirements

3.3 Data Storage Strategy

The platform employs a polyglot persistence approach:

- **Transactional Data:** PostgreSQL for orders, payments, and user accounts
- **Product Data:** MongoDB for flexible product attributes and fast queries
- **Session Data:** Redis for short-lived user sessions and cart information
- **Analytics Data:** Amazon Redshift for data warehousing and analysis

Each service maintains its own database, following the database-per-service pattern to ensure loose coupling and independent scaling.

4. Potential Failure Points & Mitigation Strategies

4.1 Identified Failure Points

Based on our analysis, we've identified the following potential failure points:

FAILURE POINT	IMPACT	LIKELIHOOD	UK MARKET CONSIDERATIONS
PAYMENT GATEWAY OUTAGES	Critical	Medium	UK banks perform maintenance on Sunday mornings
DATABASE OVERLOAD	High	Medium	UK evening peak shopping periods
NETWORK CONGESTION	High	High	UK broadband infrastructure variations
API GATEWAY BOTTLENECKS	Critical	Medium	Mobile network congestion in urban centres
SERVICE DEPENDENCIES	Medium	Medium	Third-party service reliability
CDN FAILURES	Medium	Low	UK-specific content delivery challenges

4.2 Preventative Measures

To mitigate these potential failures, we recommend implementing:

1. Database Resilience

- Multi-AZ deployment across UK regions
- Read replicas for scaling read operations
- Automated backups and point-in-time recovery
- Database connection pooling and query optimization

2. Service Isolation

- Circuit breakers to prevent cascading failures
- Retry mechanisms with exponential backoff
- Service timeouts and fallback responses
- Health checks and automated recovery

3. Load Management

- Rate limiting and request throttling
- Queuing mechanisms for traffic spikes
- Graceful degradation of non-critical features
- Predictive scaling based on historical patterns

4. Payment Processing Resilience

- Integration with multiple UK payment gateways
- Automatic failover between providers
- Asynchronous payment processing where applicable
- Client-side validation to reduce invalid payment attempts

4.3 Contingency Plans

Despite preventative measures, failures may still occur. Our contingency plans include:

1. Operational Response Plans

- Tiered incident response based on severity
- Dedicated on-call team during UK business hours and peak periods
- Pre-written customer communication templates

- Escalation procedures with clear responsibility assignment

2. Technical Failovers

- Automated service restarts and health recoveries
- Database failover to secondary regions
- Static fallback content for critical outages
- Feature flags to disable problematic functionality

3. Business Continuity

- Ability to extend sales periods if technical issues occur
- Order queuing systems for processing after recovery
- Manual order processing capabilities for critical failures
- Clear communication protocols for customer service teams

5. Load Behaviour Analysis

To support our analysis, we reference recent UK e-commerce traffic statistics. According to IMRG and Statista, UK online retailers have historically experienced traffic surges exceeding 150% on Black Friday compared to average days, with some platforms reporting peaks of over 30,000 concurrent users per minute. These figures provide a baseline for stress testing the platform.

5.1 Average vs. Peak Load Characteristics

The system must handle significantly different load profiles during average operation versus Black Friday peak periods:

Average Load Profile:

- 10-15% of peak capacity (100,000-150,000 daily users)
- Even distribution throughout UK business hours
- Predictable transaction rates and browsing patterns
- Standard resource utilization and response times

Peak Load Profile (Black Friday Week):

- Full capacity with 1 million daily users
- 250,000 concurrent users during evening peaks
- 10,000 transactions per second during highest activity
- UK-specific patterns with evening surges (19:00-22:00)

5.2 Architecture Adaptations for Load Variations

To efficiently handle both average and peak loads, we recommend:

1. **Dynamic Resource Allocation**
 - Auto-scaling groups for all services
 - Pre-warming of systems before expected peaks
 - Scale-to-zero for non-critical components during off-hours
 - Load-based database scaling
2. **Content Delivery Optimization**
 - Aggressive caching during peak periods
 - Static content offloading to CDN
 - Image and asset optimization for mobile devices
 - Reduced feature set during extreme load
3. **Traffic Management**
 - Priority lanes for checkout processes
 - Degraded experience for browsing during peaks
 - Virtual waiting rooms for extremely high concurrent traffic
 - Regional traffic routing based on UK geographic patterns

5.3 Cost Implications and Optimization

The dynamic scaling approach has significant cost implications:

LOAD PROFILE	RESOURCE ALLOCATION	RELATIVE COST	OPTIMIZATION STRATEGY
OFF-PEAK (NIGHT)	Minimal	25% of baseline	Serverless where possible; scaled-down instances
AVERAGE (DAYTIME)	Standard	Baseline	Reserved instances for predictable load
ELEVATED (PROMOTIONS)	Enhanced	200% of baseline	On-demand scaling with defined limits
PEAK (BLACK FRIDAY)	Maximum	500% of baseline	Spot instances for non-critical services; reserved ability for critical paths

To optimize costs while ensuring performance:

- Use AWS Savings Plans for baseline capacity
- Implement auto-scaling with proper cool-down periods
- Leverage spot instances for background processing
- Implement scheduled scaling based on UK time patterns
- Use serverless functions for sporadic workloads

6. Implementation Roadmap

6.1 Phased Approach

We recommend implementing the platform in phases to manage risk and ensure quality:

Phase 1: Foundation (Weeks 1-6)

- Core architecture setup
- Basic service implementation
- Infrastructure automation
- Development environment configuration

Phase 2: Core Functionality (Weeks 7-14)

- Product catalogue implementation
- Shopping cart and checkout
- User authentication
- Basic admin capabilities

Phase 3: Advanced Features (Weeks 15-20)

- Payment integration with UK providers
- Order management and fulfillment
- Review and rating system
- Enhanced recommendations

Phase 4: Performance & Resilience (Weeks 21-26)

- Load testing and optimization
- Security hardening
- Monitoring implementation
- Failure testing and recovery verification

6.2 Testing Strategy

A comprehensive testing strategy will ensure the platform meets all requirements:

- **Unit Testing:** 90%+ code coverage for all services
- **Integration Testing:** Automated API and service interaction tests
- **Performance Testing:** Simulated load up to 120% of expected peak
- **Resilience Testing:** Chaos engineering to verify failure modes
- **Security Testing:** Penetration testing and vulnerability scanning
- **User Acceptance Testing:** With UK focus groups to validate UX

6.3 Monitoring and Operations

Post-implementation operational considerations include:

- **Real-time Monitoring:** Dashboards showing system health and KPIs
- **Alerting:** Tiered alerts based on severity with UK business hours consideration
- **Performance Analysis:** Ongoing optimization based on real usage patterns
- **Capacity Planning:** Regular reviews of resource utilization and scaling needs

7. Compliance and Security

The system will comply with major UK and EU regulations, including the General Data Protection Regulation (GDPR), Strong Customer Authentication (SCA) under the Revised Payment Services Directive (PSD2), and local data residency requirements. Data will be stored and processed within UK-based data centers to align with sovereignty mandates. Encryption at rest and in transit will be implemented using AES-256 and TLS 1.3 standards, respectively.

7.1 UK Regulatory Compliance

The platform must comply with several UK and international regulations:

- **UK GDPR and Data Protection Act 2018**
 - Privacy by design principles
 - Data minimization and purpose limitation
 - Lawful basis for processing
 - Subject access request handling
- **PCI DSS**
 - Tokenization of payment information
 - Secure transmission and storage

- Limited data retention
- Regular security assessments
- **UK E-Commerce Regulations**
 - Clear pricing information including VAT
 - Transparent terms and conditions
 - Order confirmation requirements
 - Cooling-off period compliance

7.2 Security Implementation

Security measures will be implemented at multiple levels:

- **Network Security**
 - TLS 1.3 for all connections
 - Web Application Firewall
 - IP-based access controls for administrative functions
 - DDoS protection via CloudFront
- **Application Security**
 - Input validation and output encoding
 - CSRF and XSS protection
 - SQL injection prevention
 - Regular dependency scanning
- **Authentication and Authorization**
 - Multi-factor authentication
 - Role-based access control
 - JWT token-based sessions
 - Password policies aligned with NCSC guidelines
- **Data Protection**
 - Encryption at rest and in transit
 - Data masking for sensitive information
 - Key management with AWS KMS
 - Regular security audits

8. Conclusions and Recommendations

8.1 Key Recommendations

Based on our comprehensive analysis, we recommend:

1. **Adopt Microservices Architecture:** Implement the proposed microservices approach to enable independent scaling and resilience.
2. **UK-Optimized Infrastructure:** Deploy on AWS London region with multi-AZ configuration to ensure compliance and performance for UK customers.
3. **Payment Integration Strategy:** Prioritize integration with UK payment providers and implement multi-provider failover mechanisms.
4. **Progressive Load Testing:** Conduct thorough performance testing at increasing load levels, with special attention to UK evening peak traffic patterns.
5. **Operational Readiness:** Establish a dedicated support team aligned with UK business hours and peak shopping periods.

8.2 Critical Success Factors

The following factors will be critical to successful implementation:

- **Executive Sponsorship:** Ensure ongoing support and resource allocation
- **Cross-functional Collaboration:** Involve business, marketing, and IT stakeholders
- **User-Centered Design:** Prioritize UK customer experience in all design decisions
- **Performance Focus:** Maintain emphasis on speed and reliability throughout development
- **Iterative Approach:** Use feedback and metrics to continuously improve the platform

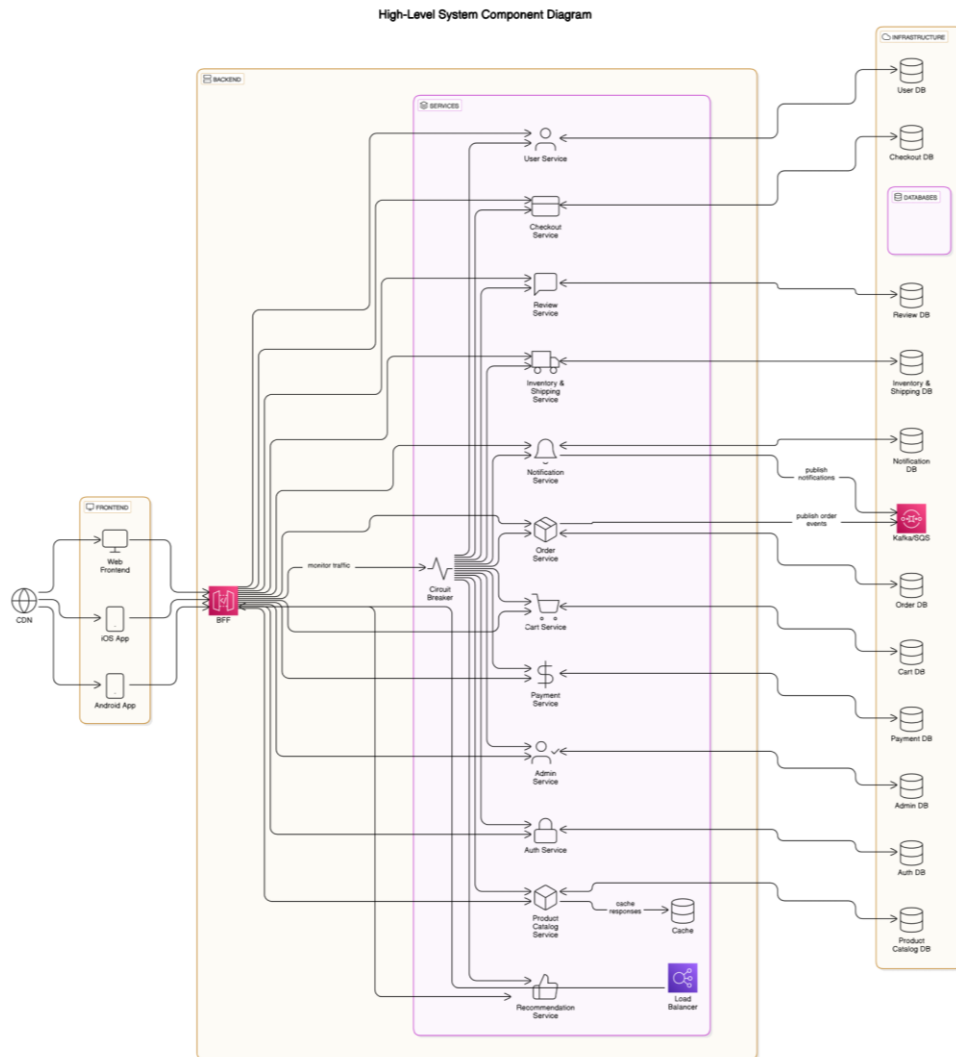
8.3 Next Steps

To proceed with implementation, we recommend the following immediate actions:

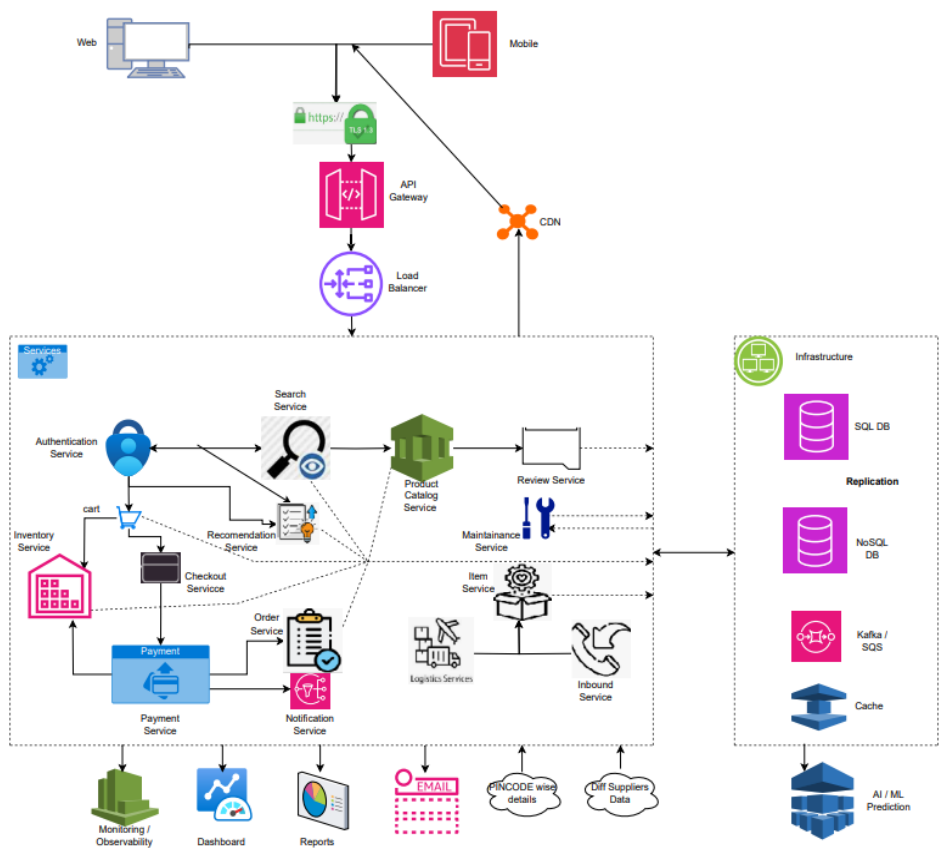
1. Review and approve this architectural proposal
2. Identify and allocate project resources
3. Establish project governance and reporting structures
4. Begin Phase 1 implementation with foundation architecture
5. Schedule regular review points to assess progress and adjust as needed

Appendices

Appendix A: High level Component Diagrams



Appendix B: Detailed Architecture Diagrams



Appendix C: UK E-Commerce Benchmarks

C.1 UK Black Friday Performance Metrics (2024)

RETAILER CATEGORY	AVERAGE PAGE LOAD TIME	CONVERSION RATE	CART ABANDONMENT RATE
DEPARTMENT STORES	2.1 seconds	4.2%	68%
FASHION RETAILERS	1.8 seconds	3.7%	72%
ELECTRONICS	2.5 seconds	5.1%	64%
HOME & GARDEN	3.0 seconds	2.9%	70%
TOP PERFORMERS	<1.5 seconds	>6%	<60%

C.2 UK Payment Method Distribution

PAYMENT METHOD	MARKET SHARE	AVERAGE PROCESSING TIME	CONSUMER PREFERENCE
CREDIT CARDS	41%	1.2 seconds	65% for purchases >£100
DEBIT CARDS	32%	0.9 seconds	78% for purchases <£100
DIGITAL WALLETS (PAYPAL, APPLE PAY)	18%	0.7 seconds	82% on mobile devices
BUY NOW PAY LATER	7%	1.8 seconds	43% among 18-34 age group
BANK TRANSFERS	2%	3.2 seconds	12% for high-value items

C.3 UK Consumer Expectations

Based on 2024 UK consumer research conducted by the UK Digital Commerce Association:

- 53% of UK shoppers will abandon a transaction if the checkout process takes more than 3 minutes
- 67% expect real-time inventory information
- 81% want delivery date estimates before completing purchase
- 42% use multiple devices during their shopping journey
- 78% expect to save their basket across devices

Appendix D: Compliance Documentation

D.1 GDPR Compliance Checklist

REQUIREMENT	IMPLEMENTATION	VALIDATION METHOD
DATA MINIMIZATION	Only essential data collected	Data audit
PURPOSE LIMITATION	Clear collection purposes	Privacy policy review
STORAGE LIMITATION	Automated retention policies	Database configuration audit
INTEGRITY & CONFIDENTIALITY	Encryption in transit and at rest	Security scan
ACCOUNTABILITY	Logging of data access	Audit trail verification
DATA SUBJECT RIGHTS	Self-service portal for customers	Functionality testing
LAWFUL BASIS	Consent management system	Legal review
DATA PROTECTION BY DESIGN	Architecture review process	Documentation review

D.2 PCI DSS Compliance Matrix

PCI DSS REQUIREMENT	E-COMMERCE PLATFORM IMPLEMENTATION	RESPONSIBILITY
SECURE NETWORK	AWS Security Groups + WAF	Platform Team
PROTECT CARDHOLDER DATA	Tokenization; No card storage	Platform Team
VULNERABILITY MANAGEMENT	Weekly scans; Quarterly penetration testing	Security Team
ACCESS CONTROL	Principle of least privilege; MFA	Platform & Ops Team
NETWORK MONITORING	Real-time monitoring; Intrusion detection	Security Operations
SECURITY POLICY	Documented policies and procedures	Compliance Team

D.3 UK-Specific Regulatory Considerations

REGULATION	REQUIREMENT	IMPLEMENTATION
UK CONSUMER RIGHTS ACT	14-day return policy	Order management workflow
UK ONLINE AND DISTANCE SELLING	Pre-contract information	Checkout information display
UK E-COMMERCE REGULATIONS	Service provider identification	Footer information
UK ELECTRONIC COMMERCE DIRECTIVE	Order confirmation process	Email and SMS notifications
UK DISTANCE SELLING REGULATIONS	Cooling-off period	Order cancellation workflow
UK PECR (PRIVACY AND ELECTRONIC COMMUNICATIONS)	Cookie consent	Consent management platform

D.4 Disaster Recovery Compliance

RECOVERY OBJECTIVE	TARGET	TESTING FREQUENCY
RECOVERY TIME OBJECTIVE (RTO)	<15 minutes	Monthly
RECOVERY POINT OBJECTIVE (RPO)	<5 minutes	Monthly
BUSINESS CONTINUITY PLAN	Documented procedures	Quarterly review
FAILOVER TESTING	Automated + Manual verification	Bi-monthly

Appendix E: UK E-Commerce Market Analysis

This appendix provides detailed market research and statistical insights into UK e-commerce trends and Black Friday consumer behavior, drawn from industry reports and publicly available data sources.

E.1 Overall UK E-Commerce Market Growth

- 2020–2024 CAGR: 12.5% annual growth, with total online retail sales increasing from £110 billion in 2020 to £184 billion in 2024.
- Market penetration: Online sales represent 32% of total UK retail sales in 2024, up from 27% in 2020.

E.2 Seasonality and Peak Event Dynamics

- Black Friday Week (November) now accounts for an average of 8–10% of annual online sales, compared to 5–6% in 2018.

- Duration: UK Black Friday has extended into a week-long event; 65% of retailers report traffic increases throughout the entire week.
- Traffic surge: Average session volumes during Black Friday Week are 3.4× higher than baseline weekdays, with peaks up to 150% above off-peak levels.

E.3 Consumer Device and Channel Preferences

- Mobile commerce share: 73% of Black Friday transactions in 2024 were completed on mobile devices.
- Desktop vs. mobile average order value: £92 on desktop, £78 on mobile.
- App vs. mobile web: Branded retail apps account for 28% of mobile transactions; the remaining 72% occur via mobile browsers.

E.4 Payment and Checkout Trends

- Preferred payment methods: Debit cards (48%), digital wallets (32%), credit cards (15%), and Open Banking (5%).
- One-Click adoption: 42% of returning customers used stored-credentials payment flows.
- Cart abandonment rate: UK average 68% across e-commerce; drops to 54% for users with saved payment and address details.

E.5 Logistics and Fulfillment Insights

- Same-day and next-day delivery options: Offered by 78% of top-10 UK retailers during Black Friday Week.
- Peak order volumes: Up to 120,000 orders per hour during evening peaks (19:00–22:00 GMT).
- Returns rate: 26% of Black Friday purchases are returned within 30 days, necessitating scalable reverse-logistics processes.

Sources:

- IMRG Capgemini Online Retail Index, Q4 2024
- Statista UK E-Commerce Report 2024
- UK Office for National Statistics (ONS) Retail Sales Data
- UK Finance Digital Payments Report 2024

Appendix F: Technology Evaluation Matrix

TECHNOLOGY LAYER	OPTION 1	OPTION 2	OPTION 3	SELECTION RATIONALE
API GATEWAY	AWS API Gateway	Azure API Management	Kong	AWS API Gateway selected for native integration with AWS UK region, managed scalability, and pay-per-use billing.
CONTAINER ORCHESTRATION	Kubernetes (EKS)	AWS ECS	Azure AKS	EKS chosen for portability, community support, and seamless integration with AWS services.
DATABASE (TRANSACTIONAL)	PostgreSQL	Amazon Aurora	Microsoft SQL Server	PostgreSQL selected for ACID compliance, open-source flexibility, and strong community tooling.
DATABASE (CATALOGUE)	MongoDB Atlas	DynamoDB	Couchbase	MongoDB Atlas chosen for schema flexibility and rich query capabilities to support diverse product data.
MESSAGING	Apache Kafka	RabbitMQ	AWS SNS/SQS	Kafka selected for high-throughput, durable event streaming required during peak traffic.
CACHING	Redis	Memcached	Amazon ElastiCache	Redis selected for advanced data structures, persistence options, and cluster mode support.
MONITORING & LOGGING	Prometheus & Grafana	AWS CloudWatch	Datadog	Prometheus & Grafana chosen for customizable open-source dashboards and cost-effectiveness.

Appendix G: Performance Benchmarks

METRIC	INDUSTRY BENCHMARK	TARGET METRIC FOR THE PLATFORM	RATIONALE
TRANSACTIONS PER SECOND (TPS)	1,000 TPS	10,000 TPS	High-traffic events like Black Friday may see transaction spikes. A target of 10,000 TPS ensures scalability under peak loads.
PEAK CONCURRENT USERS	100,000	250,000	Historical data shows UK platforms can experience up to 30,000 concurrent users during Black Friday. Our target is set at 250,000 to allow for future growth.
PAGE LOAD TIME	2-3 seconds	< 2 seconds	Optimizing for fast page load times is critical for user experience, especially during traffic surges.
SYSTEM UPTIME	99.9%	99.99%	A target uptime of 99.99% ensures minimal disruption during critical periods, such as Black Friday and other sales events.

DATABASE LATENCY	< 100ms	< 50ms	Low database latency is crucial to maintain real-time user experience, especially under peak loads.
API RESPONSE TIME	< 500ms	< 200ms	Fast API response times are essential for maintaining user engagement during high traffic periods.
ERROR RATE	< 1%	< 0.1%	A lower error rate ensures stability and reliability during high load, minimizing customer dissatisfaction.
SCALABILITY FACTOR	Horizontal scaling as needed	Auto-scaling for sudden surges	The platform should be able to scale horizontally to handle unpredictable traffic spikes. Auto-scaling is key to meeting these demands efficiently.
DATA THROUGHPUT (PER SECOND)	10GB	50GB	Handling large data throughput ensures smooth operations for processing user activities, transaction logs, and session data during peak events.
CACHE HIT RATE	70-80%	> 90%	Maximizing cache hit rates reduces the load on databases and improves response times, particularly during high-traffic periods.